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DSC630 – Predictive Analytics
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Milestone 2 – Data Selection & Project Proposal

Fake news has been around for years, but with the explosion of AI, generative content, and an insatiable appetite from the masses, it has reached critical levels and poses a severe threat to our nation. The war against fake news will be fought over various theaters, and the outcome will determine whether truth or lies triumph. Data science is essential in combating fake news, and we will aim to create a model that classifies fake news with a 75% accuracy.

This project has a few ethical concerns. Politics largely falls into left/right or Democrat/Republican paradigms. If a partisan individual analyzed this dataset, they might have an agenda to embarrass or humiliate the other side, politically speaking. As we dive into this topic, we do so in a non-partisan manner. It is unavoidable that we will encounter topics and individuals related to politics, but we will engage these topics only for the purpose of using data science tools to extract insights, train models, and tell the story.

At the conclusion of our project, we will have a model that tells people news that they read is true or false. For many, those new stories drive their beliefs and excite their emotions. We cannot delay or hide our results, regardless of which lies we expose.

There is a possibility that some of the results within the training data may have been recorded as true when they were false. The data itself originates from a fact-checking website, but truth is not as binary as a column labeled “temperature over 65.” This

distinction is crucial because the model developed from this data could be utilized to filter or silence fake news. However, if the model were manipulated or fed improper data, it could be repurposed to silence data on grounds other than its authenticity as fake news.

For my project, I've selected the Fake News dataset from Kaggle. The dataset is structured in a binary format, where true stories are represented as 1 and fake stories as 0. This problem presents two model types. Firstly, a natural language processor (NLP) will be employed to interpret the textual data. Subsequently, we will classify the data as either true or false. This model is suitable based on the data type and business objectives.

The model will be trained on a dataset comprising both genuine and fabricated data. We'll separate a test set, remove labels, and assess the percentage of false stories incorrectly classified as false. While our goal is accurate predictions, we also seek insights into fake news characteristics for detection. For example, I'd like to know if certain word combinations or key words "tip" the model off to fake news.

This task is undoubtedly fraught with risks and ethical dilemmas. To create the training data, someone had to evaluate a news story as true or false. There's a chance that a genuine news story was mistakenly recorded as false or vice versa. By making the dataset available for peer review, as Politifact and FakeNewsNet did, it becomes susceptible to criticism. Fake news is often crafted to evoke emotions and deceive readers into believing falsehoods. Consequently, individuals might develop an identity around the fake news, and

being informed that it was false could likely trigger backlash. Scientists must tread carefully on this path while upholding their ethical pursuit of truth.

There is a possibility that this dataset may not meet our criteria for an NLP model, or that our NLP model may not achieve satisfactory results. In such a scenario, we will pivot to sentiment analysis. This will provide an alternative opportunity to analyze data, create new features, and establish new prediction targets. NLPs excel at this kind of problem, but there's a possibility we switch to a different model or a evaluate a different dataset for fake news. The dataset also offers two paths for news: gossip and political. While we will focus on politics, we can pivot to using gossip news if needed.

Our overall goal is to train a model and predict unseen/test data with 75% accuracy. Secondary goals are to identify keywords for both classes for comparison and track sentiment. Also, we will conduct individual research on Twitter/X to track the headlines for some of these stories to see the type of audience/format for fake news. We've documented the ethical concerns and challenges. The data analysis will be submitted for peer reviews to ensure impartiality, quality control, and foster feedback.

Fake News Dataset: https://www.kaggle.com/datasets/algord/fake-news

Additional GitHub repo: https://github.com/KaiDMML/FakeNewsNet

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Milestone 3 – Preliminary Analysis

Since the initial proposal, I've conducted a thorough analysis for the dataset. My initial goal of training for 75% accuracy is still on target. There have been some surprises, though.

First, this problem is a bit more abstract that I realized. Text/language processing is dependent on the data source. News typically follows a format of an article title and the body or content of the article. This is extremely relevant because various fake news datasets and APIs share this common structure. However, certain models perform better at learning and predicting titles of articles instead of the article content themselves. When articles are shared on social media, the entire content is rarely read. In fact, many people don't read the entire article even if they click on it, but they're all guaranteed to read the title. My dataset contains article titles, but I could find other datasets for additional evaluation.

Although I initially and continue to utilize TextBlob for analysis and prediction, a recent assignment prompted me to alter my strategy. Upon reviewing the credit default prediction study, I gained insights into the methodologies employed by data scientists in employing multiple models and comparing classification reports to select the most effective models. Consequently, I have modified my approach to incorporate multiple

models. This not only guarantees the utilization of the optimal model but also facilitates a deeper comprehension of the data's characteristics.

Data visualization has proven instrumental in comprehending and conveying the narrative of data. For instance, plotting both the authentic and fabricated news tweet counts reveals a highly skewed distribution, predominantly on the right side. This observation stems from the presence of numerous news articles lacking tweet IDs, indicating either data omission or the non-tweeting of certain news stories. Even after standardization, the "authentic" data set exhibited a pronounced right skew due to the abundance of zeros. Histograms serve as effective tools for visualizing this distribution. Additionally, histograms were employed to illustrate sentiment, demonstrating that both datasets predominantly strive for a sentiment of zero or "neutral."

Although the article's origin is a potential feature for prediction, I have not yet utilized it as a predictive factor. This is because I intended for the content of the fabricated news to be the primary source of predictions. If time permits, I will likely incorporate this data into my overall findings, as it may enhance the accuracy of reporting fabricated news. Additionally, this data could help determine whether the primary source of fake news classification is textual content or data origin.

In summary, the original objective remains intact, but there are additional avenues for investigation. Upon attaining the 75% accuracy, we will transition to delve deeper into

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the data, thereby enabling us to present a more comprehensive and coherent picture for

the final project.